ppendix C: Thermochemical Properties of Combustion Products and Other Ideal Gases

Table C-1
Thermochemical Properties of N_2 and N

	Nitrogen, Diatomic (N_2) $\Delta\bar{h}^o_{298} = 0$ kJ/kmol $M = 28.013$		Nitrogen, Monatomic (N) $\Delta\bar{h}^o_{298} = 472\ 680$ kJ/kmol $M = 14.007$	
T K	$\left(\bar{h} - \bar{h}^o_{298}\right)$ kJ/kmol	log K_p	$\left(\bar{h} - \bar{h}^o_{298}\right)$ J/kmol	log K_p $1/2N_2 \leftrightarrow N$
0	–8670	0.000	–6197	Infinite
100	–5768	0.000	–4119	-243.583
200	–2857	0.000	–2040	-120.405
298	0	0.000	0	-79.800
300	54	0.000	38	-79.289
400	2971	0.000	2117	-58.704
500	5911	0.000	4196	-46.336
600	8894	0.000	6274	-38.081
700	11937	0.000	8353	-32.177
800	15046	0.000	10431	-27.744
900	18223	0.000	12510	-24.292
1000	21463	0.000	14589	-21.528
1100	24760	0.000	16667	-19.265
1200	28109	0.000	18746	-17.377
1300	31503	0.000	20825	-15.778
1400	34936	0.000	22903	-14.406
1500	38405	0.000	24982	-13.217
1600	41904	0.000	27060	-12.175
1700	45430	0.000	29139	-11.256
1800	48979	0.000	31218	-10.437
1900	52549	0.000	33296	-9.705
2000	56137	0.000	35375	-9.046
2100	59748	0.000	37455	-8.449
2200	63362	0.000	39534	-7.905
2300	67007	0.000	41614	-7.409
2400	70640	0.000	43695	-6.954
2500	74312	0.000	45777	-6.535
2600	77963	0.000	47860	-6.149
2700	81659	0.000	49949	-5.790
2800	85323	0.000	52033	-5.457
2900	89035	0.000	541 24	-5.147
3000	92715	0.000	56218	-4.858
3200	100134	0.000	60420	-4.332
3400	107577	0.000	64646	-3.868
3600	115042	0.000	68902	-3.455
3800	122526	0.000	73194	-3.086
4000	130027	0.000	77532	-2.752

Source: Sontag, R. E., Borgnakke, C., and Van Wylen, G. J., *Fundamentals of Thermodynamics*, John Wiley & Sons, Inc, 1999, and Keating, E. L., *Applied Combustion*, CRC Press, 2007

Table C-2

Thermochemical Properties of CO_2 and CO

	Carbon Dioxide (CO_2) $\Delta\bar{h}^o_{298} = -393\ 522$ kJ/kmol $M = 44.01$		Carbon Monoxide (CO) $\Delta\bar{h}^o_{298} = -110\ 527$ kJ/kmol $M = 28.01$	
T K	$\left(\bar{h} - \bar{h}^o_{298}\right)$ kJ/kmol	log K_p $C+O_2 \leftrightarrow CO_2$	$\left(\bar{h} - \bar{h}^o_{298}\right)$ kJ/kmol	log K_p $C + 1/2O_2 \leftrightarrow CO$
0	-9364	Infinite	-8671	Infinite
100	-6457	205.645	-5772	62.809
200	-3413	102.922	-2860	33.566
298	0	69.095	0	24.029
300	69	68.670	54	23.910
400	4003	51.540	2977	19.109
500	8305	41.260	5932	16.235
600	12906	34.405	8942	14.318
700	17754	29.506	12021	12.946
800	22806	25.830	15174	11.914
900	28030	22.970	18397	11.108
1000	33397	20.680	21686	10.459
1100	38885	18.806	25031	9.926
1200	44473	17.243	28427	9.479
1300	50148	15.920	31867	9.099
1400	55895	14.785	35343	8.771
1500	61705	13.801	38852	8.485
1600	67569	12.940	42388	8.234
1700	73480	12.180	45948	8.011
1800	79432	11.504	49529	7.811
1900	85420	10.898	53128	7.631
2000	91439	10.353	56743	7.469
2100	97500	9.860	60375	7.321
2200	103562	9.411	64012	7.185
2300	109671	9.001	67676	7.061
2400	115779	8.625	71326	6.946
2500	121926	8.280	75023	6.840
2600	128074	7.960	78679	6.741
2700	134256	7.664	82408	6.649
2800	140435	7.388	86070	6.563
2900	146645	7.132	89826	6.483
3000	152853	6.892	93504	6.407
3200	165321	6.458	100962	6.269
3400	177836	6.074	108440	6.145
3600	190394	5.732	115938	6.034
3800	202990	5.425	123454	5.933
4000	215624	5.149	130989	5.841

Source: Sontag, R. E., Borgnakke, C., and Van Wylen, G. J., *Fundamentals of Thermodynamics*, John Wiley & Sons, Inc, 1999, and Keating, E. L., *Applied Combustion*, CRC Press, 2007

Table C-3
Thermochemical Properties of O_2 and O

T K	Oxygen, Diatomic (O_2) $\Delta\bar{h}^o_{298} = 0$ kJ/kmol $M = 31.999$ $\left(\bar{h} - \bar{h}^o_{298}\right)$ kJ/kmol	log K_p	Oxygen, Monatomic (O) $\Delta\bar{h}^o_{298} = -249\ 170$ kJ/kmol $M = 16.00$ $\left(\bar{h} - \bar{h}^o_{298}\right)$ kJ/kmol	log K_p $1/2O_2 \leftrightarrow O$
0	-8683	0.000	-6725	Infinite
100	-5777	0.000	-4518	-126.730
200	-2868	0.000	-2186	-61.992
298	0	0.000	0	-40.604
300	54	0.000	41	-40.334
400	3027	0.000	2207	-29.473
500	6086	0.000	4343	-22.940
600	9245	0.000	6462	-18.574
700	12499	0.000	8570	-15.449
800	15836	0.000	10671	-13.101
900	19241	0.000	12767	-11.272
1000	22703	0.000	14860	-9.807
1100	26212	0.000	16950	-8.606
1200	29761	0.000	19039	-7.604
1300	33345	0.000	21126	-6.755
1400	36958	0.000	23212	-6.027
1500	40600	0.000	25296	-5.395
1600	44267	0.000	27381	-4.842
1700	47959	0.000	29464	-4.353
1800	51674	0.000	31547	-3.918
1900	55414	0.000	33630	-3.529
2000	59176	0.000	35713	-3.178
2100	62986	0.000	37798	-2.860
2200	66770	0.000	39878	-2.571
2300	70634	0.000	41961	-2.307
2400	74453	0.000	44045	-2.065
2500	78375	0.000	46133	-1.842
2600	82225	0.000	48216	-1.636
2700	86199	0.000	50304	-1.446
2800	90080	0.000	52391	-1.268
2900	94111	0.000	54484	-1.103
3000	98013	0.000	56574	-0.949
3200	106022	0.000	60767	-0.670
3400	114101	0.000	64971	-0.423
3600	122245	0.000	69190	-0.204
3800	130447	0.000	73424	-0.007
4000	138705	0.000	77675	0.170

Source: Sontag, R. E., Borgnakke, C., and Van Wylen, G. J., *Fundamentals of Thermodynamics*, John Wiley & Sons, Inc, 1999, and Keating, E. L., *Applied Combustion*, CRC Press, 2007

Table C-4
Thermochemical Properties of H_2 and H

	Hydrogen (H_2) $\Delta\bar{h}^o_{298} = 0$ kJ/kmol $M = 2.016$		Hydrogen, Monatomic (H) $\Delta\bar{h}^o_{298} = 217\,999$ kJ/kmol $M = 1.008$	
T K	$\left(\bar{h} - \bar{h}^o_{298}\right)$ kJ/kmol	log K_p	$\left(\bar{h} - \bar{h}^o_{298}\right)$ kJ/kmol	log K_p $1/2H_2 \leftrightarrow H$
0	–8467	0.000	–6197	Infinite
100	–5467	0.000	–4119	-110.954
200	–2774	0.000	–2040	-54.322
298	0	0.000	0	-35.612
300	53	0.000	38	-35.377
400	2961	0.000	2117	-25.876
500	5883	0.000	4196	-20.158
600	8799	0.000	6274	-16.336
700	11730	0.000	8353	-13.599
800	14681	0.000	10431	-11.539
900	17657	0.000	12510	-9.934
1000	20663	0.000	14589	-8.646
1100	23704	0.000	16667	-7.589
1200	26785	0.000	18746	-6.707
1300	29907	0.000	20825	-5.958
1400	33073	0.000	22903	-5.315
1500	36281	0.000	22982	-4.756
1600	39533	0.000	24060	-4.266
1700	42826	0.000	29139	-3.833
1800	46160	0.000	31218	-3.448
1900	49532	0.000	33296	-3.102
2000	52942	0.000	35375	-2.790
2100	56379	0.000	37455	-2.508
2200	59865	0.000	39532	-2.251
2300	63371	0.000	41610	-2.016
2400	66915	0.000	43689	-1.800
2500	70492	0.000	45769	-1.601
2600	74082	0.000	47847	-1.417
2700	77718	0.000	49928	-1.247
2800	81355	0.000	52004	-1.089
2900	85044	0.000	54082	-0.941
3000	88725	0.000	56161	-0.803
3200	96187	0.000	60318	-0.553
3400	103736	0.000	64475	-0.332
3600	111367	0.000	68633	-0.135
3800	119077	0.000	72790	0.042
4000	126864	0.000	76947	0.201

Source: Sontag, R. E., Borgnakke, C., and Van Wylen, G. J., *Fundamentals of Thermodynamics*, John Wiley & Sons, Inc, 1999, and Keating, E. L., *Applied Combustion*, CRC Press, 2007

Table C-5
Thermochemical Properties of NO and NO_2

T K	Nitric Oxide (NO) $\Delta\bar{h}^o_{298}$ = 90 291 kJ/kmol, M = 30.006: $\left(\bar{h}-\bar{h}^o_{298}\right)$ kJ/kmol	log K_p $1/2N_2+1/2O_2 \leftrightarrow NO$	Nitrogen, Dioxide (NO_2) $\Delta\bar{h}^o_{298}$ = 33 100 kJ/kmol, M = 46.005: $\left(\bar{h}-\bar{h}^o_{298}\right)$ kJ/kmol	log K_p $1/2N_2+O_2 \leftrightarrow NO_2$
0	–9192	Infinite	–10186	Infinite
100	–6073	-46.453	–6861	-20.859
200	–2951	-22.929	–3495	-11.859
298	0	-15.171	0	-8.977
300	55	-15.073	68	-8.941
400	3040	-11.142	3927	-7.513
500	6059	-8.783	8099	-6.669
600	9144	-7.210	12555	-6.111
700	12308	-6.086	17250	-5.714
800	15548	-5.243	22138	-5.417
900	18858	-4.587	27180	-5.185
1000	22229	-4.062	32344	-5.000
1100	25653	-3.633	37606	-4.848
1200	29120	-3.275	42946	-4.721
1300	32626	-2.972	48351	-4.612
1400	36164	-2.712	53808	-4.519
1500	39729	-2.487	59309	-4.438
1600	43319	-2.290	64846	-4.367
1700	46929	-2.116	70414	-4.304
1800	50557	-1.962	76008	-4.248
1900	54201	-1.823	81624	-4.198
2000	57859	-1.699	87259	-4.152
2100	61530	-1.586	92914	-4.111
2200	65212	-1.484	98578	-4.074
2300	68906	-1.391	104261	-4.040
2400	72606	-1.305	109948	-4.008
2500	76320	-1.227	115650	-3.979
2600	80034	-1.164	121358	-3.953
2700	83764	-1.087	127081	-3.928
2800	87491	-1.025	132800	-3.905
2900	91232	-0.967	138536	-3.884
3000	94973	-0.913	144267	-3.864
3200	102477	-0.815	155756	-3.828
3400	110000	-0.729	167262	-3.797
3600	117541	-0.653	178783	-3.770
3800	125099	-0.585	190316	-3.746
4000	132671	-0.524	201860	-3.724

Source: Sontag, R. E., Borgnakke, C., and Van Wylen, G. J., *Fundamentals of Thermodynamics*, John Wiley & Sons, Inc, 1999, and Keating, E. L., *Applied Combustion*, CRC Press, 2007

Table C-6
Thermochemical Properties of H_2O and OH

	Water (H_2O) $\Delta\bar{h}^o_{298} = -241\ 826$ kJ/kmol $M = 18.015$		Hydroxyl (OH) $\Delta\bar{h}^o_{298} = 38\ 987$ kJ/kmol $M = 17.007$	
T K	$\left(\bar{h} - \bar{h}^o_{298}\right)$ kJ/kmol	log K_p $H_2 + 1/2O_2 \leftrightarrow H_2O$	$\left(\bar{h} - \bar{h}^o_{298}\right)$ kJ/kmol	log K_p $1/2H_2 + 1/2O_2 \leftrightarrow OH$
0	-9904	Infinite	-9172	Infinite
100	-6617	123.600	-6140	-19.672
200	-3282	60.792	-2975	-9.475
298	0	40.048	0	-6.089
300	62	39.786	55	-6.046
400	3450	29.240	3034	-4.327
500	6922	22.886	5991	-3.296
600	10499	18.633	8943	-2.609
700	14190	15.583	11902	-2.120
800	18002	13.289	14881	-1.755
900	21937	11.498	17889	-1.472
1000	26000	10.062	20935	-1.247
1100	30190	8.883	24024	-1.064
1200	34506	7.899	27159	-0.912
1300	38941	7.064	30340	-0.784
1400	43491	6.347	33567	-0.674
1500	48149	5.725	36838	-0.580
1600	52907	5.180	40151	-0.497
1700	57757	4.699	43502	-0.425
1800	62693	4.270	46890	-0.361
1900	67706	3.886	50311	-0.304
2000	72788	3.540	53763	-0.253
2100	77831	3.227	57241	-0.207
2200	83153	2.942	60751	-0.165
2300	88295	2.682	64283	-0.127
2400	93741	2.443	67840	-0.092
2500	98964	2.224	71417	-0.060
2600	104520	2.021	75018	-0.031
2700	109813	1.833	78634	-0.004
2800	115463	1.658	82268	0.021
2900	120813	1.495	85918	0.044
3000	126548	1.343	89585	0.065
3200	137756	1.067	96960	0.104
3400	149073	0.824	104388	0.137
3600	160484	0.607	111864	0.167
3800	171981	0.413	119382	0.193
4000	183552	0.238	126940	0.216

Source: Sontag, R. E., Borgnakke, C., and Van Wylen, G. J., *Fundamentals of Thermodynamics*, John Wiley & Sons, Inc, 1999, and Keating, E. L., *Applied Combustion*, Second Edition, CRC Press, 2007

...nthalpies and Free Energies of Formation of Several Compounds at Standard State of 1 bar and 25°C

Compound (phase)	Formula	M (kg/kmol)	h^f (MJ/kmol)	g^f (MJ/kmol)
Acetylene (g)	C_2H_2	26.04	226.75	209.20
Ammonia (g)	NH_3	17.04	−46.19	−16.64
Benzene (g)	C_6H_6	78.11	82.93	129.66
Carbon-graphite (s)	C	12.01	0.00	0.00
Carbon-diamond (s)	C	12.01	1.88	2.89
Carbon dioxide (g)	CO_2	44.01	−393.52	−394.36
Carbon monoxide (g)	CO	28.01	−110.53	−137.15
Ethane (g)	C_2H_6	30.07	−84.68	−32.89
Ethanol (g)	C_2H_5OH	46.07	−235.31	−168.57
Ethene (g)	C_2H_4	28.05	52.28	68.12
Hydrogen (g)	H_2	2.02	0.00	0.00
Hydrogen peroxide (l)	H_2O_2	34.02	−187.61	−117.99
Hydrogen sulfide (l)	H_2S	34.00	−20.15	−33.02
Methane (g)	CH_4	16.04	−74.85	−50.69
Methanol (g)	CH_3OH	32.05	−200.67	−162.00
Nitrogen (g)	N_2	28.01	0.00	0.00
Nitrogen atomic (g)	N	14.01	472.65	455.50
Nitric oxide	NO	30.01	90.42	86.72
Nitrogen dioxide	NO_2	46.05	33.32	51.32
Nitrous oxide	N_2O	44.02	81.84	103.90
n-Butane (g)	C_4H_{10}	58.12	−124.73	−15.71
n-Heptane (g)	C_7H_{16}	100.20	−187.82	−8.11
n-Hexane (g)	C_6H_{14}	86.17	−167.20	−0.29
n-Octane (l)	C_8H_{18}	114.22	. 209.45	16.53
n-Pentane (l)	C_5H_{12}	72.15	−146.44	−8.20
Oxygen (g)	O_2	32.00	0.00	0.00
Oxygen atomic(g)	O	16.00	249.20	231.76
Ozone (g)	O_3	48.00	34.00	39.06
Propane (g)	C_3H_8	44.09	−103.85	−23.49
Propene (g)	C_3H_6	42.08	20.41	62.72
Sulfur rhombic (s)	S	32.00	0.00	0.00
Sulfur monoclinic (s)	S	32.00	0.30	0.10
Sulfur dioxide (g)	SO_2	64.07	−296.90	−300.37
Sulfur trioxide (g)	SO_3	80.06	−395.18	−370.37
Water (l)	H_2O	18.02	−285.84	−237.19
Water (g)	H_2O	18.02	−241.83	−228.60

The enthalpy change, ΔH, in the reaction chamber may be br...
follows: